

Exercise 1. PCA and classification. Consider the data set `exercise_21_01_26.npy` that contains $N = 20000$ observations from a binary classification problem. The data set contains feature-label observations (X_i, Y_i) , where $X_i \in \mathbb{R}^{10}$ and $Y_i \in \{-1, 1\}$.

- Reduce the dimensionality of the data set to 2 components. Assess the goodness of such a dimensionality reduction by examining the explained variance of the principal components.
- Plot the data in the PCA domain and comment on the separability of the classification problem. Identify the appropriate decision statistic and estimate the parameters of the corresponding supervised classifier by using the SGD algorithm. Implement the training phase over 75% of the observations, and the testing phase over the remaining observations.
- Consider the same training and test sets used in the previous point. Postulate an analytical form for the data distribution (conditioned on the labels), and use the training data to estimate the parameters of the proposed likelihood model. Then, derive and implement the MAP classifier.
- Compare the test error probability of the supervised classifier against the error probability of the MAP classifier.

Exercise 2. Online supervised learning. The file `online_learning.npz` contains data collected from an online status reporting system. For each time instant $i = 1, 2, \dots, N$, the system collects features $X_i \in \mathbb{R}^2$ and the corresponding label $Y_i \in \{\text{working}, \text{not working}\}$. A classifier must be implemented to predict whether the system is working or not. Accordingly, the file `online_learning.npz` contains: a) a training set consisting of a stream of $N_{\text{train}} = 16000$ feature-label pairs (X_i, Y_i) , and b) a test set with $N_{\text{test}} = 4000$ samples. At time $i = 8000$, the distribution of the features changes, reflecting a shift in the system behavior.

- Plot, separately, the first part of the data up to time $i = 8000$ and the second part of the data from $i = 8001$ onwards. Comment on how the feature distribution changes.
- Postulate an analytical form for the likelihood models in each of the aforementioned data chunks, and use the training data to estimate the parameters of these models. Use the resulting likelihoods to derive the two MAP classifiers corresponding to the two chunks.
- Suppose that, as the samples of the training data stream arrive, a logistic classifier is trained by using the SGD algorithm with constant step-size. Implement this training strategy **over the entire training set** (without reset/restart after the change) and comment on the evolution/adaptation of the estimated parameters in view of the aforementioned distribution change.
- Compute and compare the test error probabilities of the two MAP classifiers and the two logistic classifiers. Assume that the system designer knows that the first half of the available test samples is related to the first data distribution, while the second half is related to the second distribution.

Exercise 3. Binary classification with different decision statistics. Consider a binary classification problem where $X = [X_1, X_2] \in \mathbb{R}^2$ is a feature vector and $Y \in \{-1, 1\}$ is the hypothesis. The likelihoods of the features (which are assumed to be independent given the hypothesis) are the following:

$$\begin{aligned} \ell(x_1|Y = +1) &= \frac{1}{\sqrt{3\pi}} \exp\left\{-\frac{(x_1 - 1)^2}{3}\right\}, & \ell(x_2|Y = +1) &= \frac{1}{\sqrt{3\pi}} \exp\left\{-\frac{(x_2 - 1)^2}{3}\right\}, \\ \ell(x_1|Y = -1) &= \frac{1}{\sqrt{0.6\pi}} \exp\left\{-\frac{(x_1 - 1.5)^2}{0.6}\right\}, & \ell(x_2|Y = -1) &= \frac{1}{\sqrt{0.6\pi}} \exp\left\{-\frac{(x_2 + 1.5)^2}{0.6}\right\}, \end{aligned}$$

and the prior probabilities for the hypotheses are uniform.

- Generate a training set for the assigned problem with $N_{\text{train}} = 15000$ samples, and a test set with $N_{\text{test}} = 5000$ samples.
- Derive the decision rule for the optimal classifier.
- Using the SGD algorithm, learn the parameters of two supervised classifiers, one using a linear decision statistic, the other using a quadratic decision statistic. **Note: choose carefully the form of the quadratic decision statistic by visually examining the data set. Verify that the quadratic decision statistic with the learned parameters approximates well the optimal decision statistic computed in the previous point.**
- Compute the accuracy, on the test set, of the two classifiers obtained in the previous point and of the optimal classifier in point b. Comment on the results.